Development of Electrostatic Levitation Motor for Vacuum Condition

Akio YAMAMOTO[1], Hidehiko YASUI[2], Naoshige SHIMIZU[3], and Toshiro HIGUCHI[4]

[1,2,3,4]The University of Tokyo, Tokyo, Japan,
e-mail : akio@ieee.org[1], (hide[2], naoshige[3], higuchi[4])@intellect.pe.u-tokyo.ac.jp

Abstract— **To drive a motor in vacuum condition, many factors must be considered, such as friction and heat generation. Since typical electromagnetic motors structurally have problems of friction and heat generation, it is difficult to use them in vacuum condition. Besides, in some vacuum applications like EB lithography, SEM, and TEM, electromagnetic motors are unlikely to be used since existence of electromagnetic field is a fatal problem. We proposes an application of non-contact levitation electrostatic motor using *Dual Excitation Multiphase Electrostatic Drive* (DEMED) system to those fields.**

We have developed a direct drive high-power electrostatic linear motor called DEMED. DEMED is a linear synchronous motor that consists of pairs of thin plates, stator and slider, and possesses a power per weight ratio of 230W/kg and acceleration of over 100G in an open loop systems in dielectric liquid. These performances are higher than those of typical electromagnetic drives.

In this paper non-contact levitation and driving control system for DEMED is proposed. A levitation motor using DEMED utilizes the same electrodes for both levitation and driving. For controlling attractive force, bias voltage is superposed onto driving voltage. We have fabricated prototype motor with 2-DOF and have realized levitation drive control. The fabricated motor was operated at a speed of 20mm/s with keeping a levitation gap of 200μm.

Index Terms— **Electrostatic motor, Levitation motor, Vacuum, Electron beam**

I. Introduction

In the fields of semiconductor manufacturing and micro/nano structure fabrication process, there is a strong demand for clean environment. By removing mechanical contact, a clean environment can be achieved. In order to achieve it, levitating mechanism is necessary to remove mechanical contact. In the past, magnetic bearing and magnetic levitation drive were proposed to achieve clean environment [1–3].

However, in recent years, to create a clean environment is no longer enough to meet the manufacturing requirements. For example, electron beam projection lithography(EPL) is one of several promising technologies, which need vacuum condition, as well as extremely low surrounding electromagnetic field. [4–11]. In this case, actuators such as magnetic bearing and magnetic levitation, which utilize magnetic field for their force generation, cannot be used in the mentioned process [6]. This is because existence of electromagnetic field in the process is a fatal problem.

In this paper, we focus on the use of electrostatic actuator, instead of electromagnetic field drive actuator. Electrostatic actuator is driven with high voltage and low current that generates a very small magnetic field. Generally, electrostatic actuators have been considered suitable only for driving micro-machines. This is because the scale effect of electrostatic force affirms that force per volume ratios of electrostatic actuators increase as their dimensions decrease [12, 13]. Therefore many studies on micro electrostatic actuators have been carried out [14–17]. However, Niino et al. revealed that electrostatic actuator could be strong enough even in centimeter dimensions in the study on *Dual Excitation Multiphase Electrostatic Drive (DEMED)* [18, 19]. In this paper, we propose electrostatic levitation motor based on this actuator.

Several attempts have been reported on electrostatic levitation system. Knoebel showed the electrostatic vacuum gyro, which used electrostatic force for levitation and electromagnetic force for driving [20]. Apparently, this system is not suitable for the mentioned applications. Kumar et al. proposed electrostatic levitating micro-motors, on which both levitation and driving are realized using electrostatic force [21]. In this system, they realized driving and levitation on separate electrodes. In order to achieve higher thrust force, larger electrodes' area is needed because generation of electrostatic force is in proportion to the area. However, as space in chamber is limited and valuable, a smaller actuator with high output thrust force is desired. From this point of view, to realize driving and levitation on separates electrodes is not efficient.

In this paper, we propose new electrostatic levitation motor system that realizes driving and levitation on the same electrodes. The proposed system adopts *Dual Excitation Multiphase Electrostatic Drive (DEMED)*, which is proven to be able to generate high thrust force, for its driving principle.

II. Dual Excitation Multiphase Electrostatic Drive (DEMED)

This section overviews principle of DEMED and major performances obtained in past research. A basic structure of DEMED is shown in Fig. 1. DEMED consists of a pair of flat films, slider and stator. Both films are made of polyimide and contain copper electrodes. The electrodes are aligned with a constant pitch. Electrodes on slider and stator are connected into three phases, respectively. The shape of electrodes is straight in stator, and washboard-shaped in slider [22].

For driving, three-phase ac voltage is applied to both the three electrodes of slider and stator. The connections of three-phase ac voltages are reversed on slider and on

0-7803-7912-8/03/$17.00 © 2003 IEEE

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.

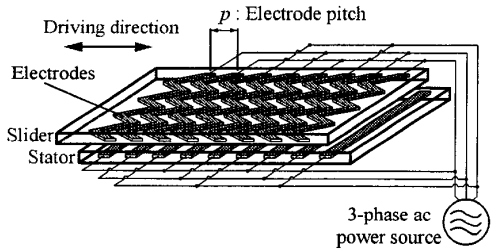


Fig. 1. Schematic illustration of Dual Excitation Multiphase Electrostatic Drive

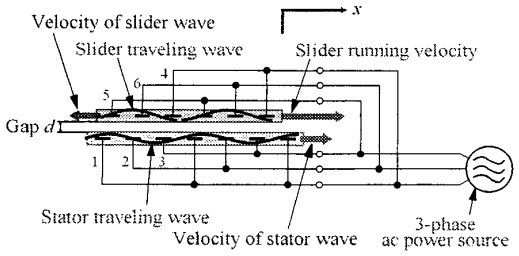


Fig. 2. Principle of driving DEMED

stator. Applying ac voltage causes traveling potential waves on slider and stator as shown in Fig. 2. The electrostatic interaction between two potential waves gives thrust force to the slider. The velocity of the slider is expressed as $v = 6pf$, where v is slider velocity, p is electrode pitch, f is frequency of applied three-phase ac voltage. In closed loop control system, applied voltage $\boldsymbol{V}$ is controlled as

$$\boldsymbol{V} = \Big[V_0\sin(\phi), V_0\sin(\phi - \frac{2}{3}\pi), V_0\sin(\phi + \frac{2}{3}\pi) \quad V_0\sin(\phi), V_0\sin(\phi + \frac{2}{3}\pi), V_0\sin(\phi - \frac{2}{3}\pi)\Big]^t. \quad (1)$$

where V_0 is the amplitude of input voltage. Phase ϕ is expressed as

$$\phi = -\frac{\pi}{3p}x - \frac{\pi}{6} \pm \frac{\pi}{4}, \quad (2)$$

where x is slider position [23]. Then, thrust force f_x and attractive force f_d are expressed as

$$\begin{cases} f_x = \pm\kappa_x \dfrac{V_0^2}{d} \\ f_d = \kappa_a \dfrac{V_0^2}{d^2}, \end{cases} \quad (3)$$

where κ_x and κ_a are determined by the form factor of the motor.

III. Proposal of Electrostatic Levitation Motor

A. Force generation

In this paper, we propose a new electrostatic levitation motor that realizes driving and levitation using the same

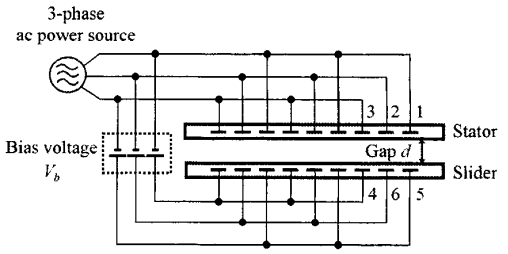


Fig. 3. Schematic illustration of independent control system

electrodes. Fig. 3 shows schematic illustration of the system. This system utilizes the same driving principle as DEMED, however, slider is suspended below stator. To realize electrostatic levitation motor, it is necessary to control thrust and attractive force independently. However equations (3) states that thrust force f_x and attractive force f_d are always proportional to each other. Therefore they cannot be controlled independently.

We have revealed that applying bias voltage to slider electrodes enables us to control attractive force independently from thrust force [23]. When bias voltage V_b is applied to slider electrodes, input voltage $\boldsymbol{V}$ is expressed as

$$\boldsymbol{V} = \Big[V_0\sin(\phi), V_0\sin(\phi - \frac{2}{3}\pi), V_0\sin(\phi + \frac{2}{3}\pi) \quad V_0\sin(\phi) + V_b, V_0\sin(\phi + \frac{2}{3}\pi) + V_b, V_0\sin(\phi - \frac{2}{3}\pi) + V_b\Big]^t, \quad (4)$$

and thrust force f_x and attractive force f_d can be rewritten as

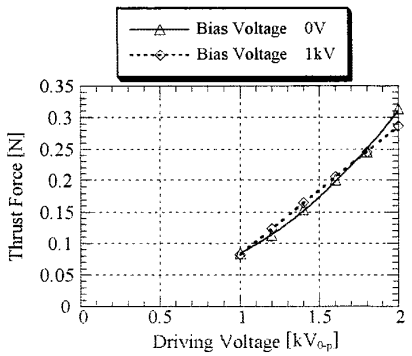
$$\begin{cases} f_x = \pm\kappa_x \dfrac{V_0^2}{d} \\ f_d = \dfrac{1}{d^2}(\kappa_a V_0^2 + \kappa_b V_b^2), \end{cases} \quad (5)$$

where κ_b is derived from the form factor of motor. Equations (5) show that bias voltage does not influence thrust force. Therefore, by using bias voltage to compensate the change of V_0, attractive force becomes independent from thrust force.

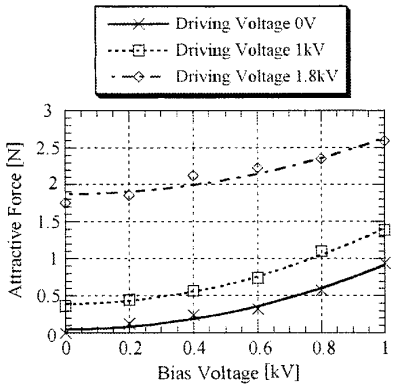
B. Measurement of force

We verified (5) experimentally. The forces, f_x and f_d, were measured using universal force-moment sensor system (UFS-20A05, NITTA Co.). Results are plotted in Fig. 4. For measurement, DEMED with electrode pitch of 500μm was used. The measurement was carried out with the gap d of 200μm. Fig. 4(a) shows relationship between thrust force and driving voltage with different bias voltage. It is verified thrust force is not affected by bias voltage. Fig. 4(b) shows relationship between attractive force and bias voltage with different driving voltage. Attractive force increases in accordance with driving voltage. These results agree with (5) very well.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.



(a) Variation of thrust force as a function of driving voltage with different bias voltage



(b) Variation of attractive force as a function of bias voltage with different driving voltage

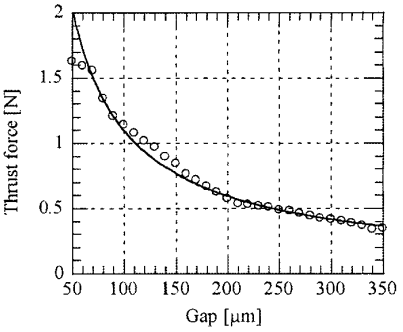
Fig. 4. Thrust force and attractive force against voltages

IV. Relationship between gap and forces

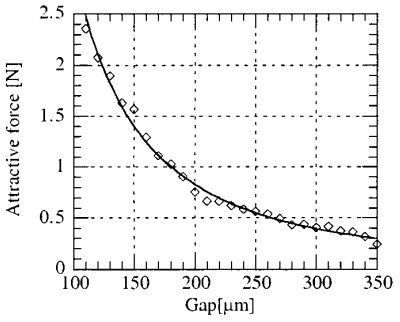
To realize levitation motor, it is necessary to know how thrust force and attractive force vary in accordance with gap. We measured the relationship among thrust force, attractive force, and gap, at an applied voltage V_0 of 3kV. Fig. 5 shows the result of the measurement between forces and gap. Fig. 5(a) and (b) show that thrust and attractive forces are inversely proportional to square of the gap.

V. Fabrication of prototype

This section describes fabrication of electrostatic levitation motor with linear sliding unit. To verify the proposed control theory, we have fabricated an experimental prototype with linear sliding unit. Fig. 6(a) and (b) show the setup. This setup has two degree of freedom, one is for driving direction and the other is for rotational direction around linear sliding unit. The rotational direction is used as vertical motion of slider. Slider is suspended by the attractive force at a disired angle. Since minute rotatinal motion can be approximated by vertical translation, we will deal with the motion as vertical translation in the following discussion. Counter-balance weight is installed on the slider to reduce effective mass of the slider, for ease of experiments. Slider's



(a) Thrust force vs. gap



(b) Attractive force vs. gap

Fig. 5. Relationship between forces and gap

weight is approximately 350g including the counter mass.

On the surfaces of slider and stator, flexible printed circuit films that have electrodes as in Fig. 6(c) are attached. Center electrodes are used as position sensor and side electrodes as gap sensor. Pitch of electrodes for driving and levitation is 500μm, same applies to those for position sensor. The shape of electrodes is straight in stator and washboard-shaped in slider.

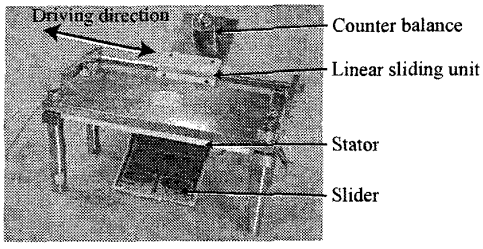
A. Gap sensor

The system has built-in gap sensor as shown in Fig. 6(c). The circuit diagram of the gap sensor is shown in Fig. 7. The gap sensor consists of capacitance bridge circuit, in which C_g expresses capacitance between electrodes installed on stator and slider. Capacitance C_v is variable capacitance for balancing the bridge. Output voltage of the bridge circuit is expressed as

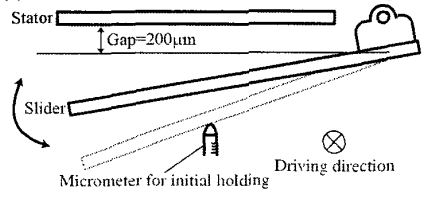
$$V_{out} = \frac{A_g C_0 (C_v - C_g)}{2(C_0 + C_g)(C_0 + C_v)}. \quad (6)$$

In our implementation, C_0 is 50pF. The output of the bridge is fed into synchronous detection circuit to detect variation of C_g, which varies with gap between stator and slider. Since electrodes for driving and gap sensor is quite near, stray capacitance connects electrodes for those. When driving voltage of 1-2kV is applied, the output of bridge is affected by driving voltage, and could become several hundred volt. Since OP-amp circuit cannot handle such high voltage, passive high-pass filters with sufficiently high impedance are installed before detection circuit. Frequency and amplitude of the input voltage was 100kHz and $20V_{p-p}$. The sensor

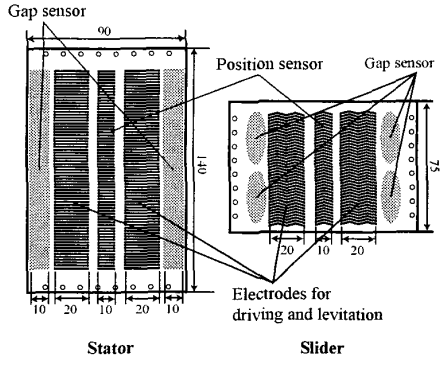
Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.



(a) Electrostatic levitation motor with linear sliding unit



(b) Illustration of electrostatic levitation motor with linear sliding unit



(c) Electrode films

Fig. 6. Experimental setup

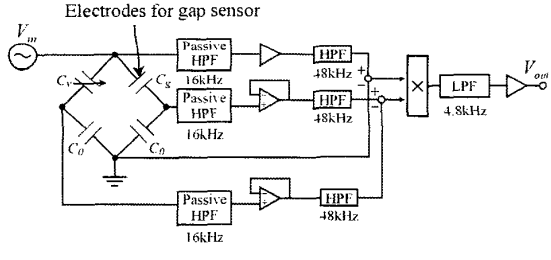


Fig. 7. Circuit diagram of gap sensor

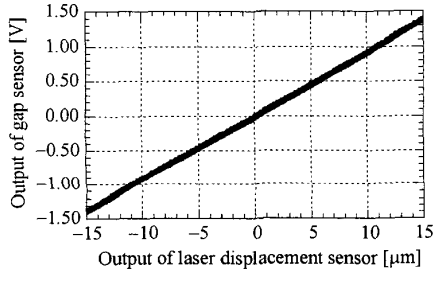


Fig. 8. Output voltage of gap sensor

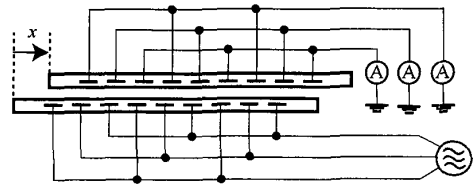


Fig. 9. Principle of position sensor

output is plotted in Fig. 8. Hysteresis and non-linearity are hardly seen. Obtained resolution is 0.16μm. This performance is sufficient to carry out actual controlling.

B. Position sensor

Position sensor utilizes the center electrodes in Fig. 6(c). Schematic diagram of position sensor is shown in Fig. 9 [23]. The electrodes have the same structure as driving electrodes. When a three-phase ac voltage expressed by $\boldsymbol{V_s}$ is applied to the stator's three-phase electrodes, electric currents $\boldsymbol{I}$ flow into slider's electrodes. Let $\boldsymbol{V_s}$ be

$$\boldsymbol{V_s} = [V_a \sin(\omega_s t), V_a \sin(\omega_s t + \frac{2\pi}{3}), V_a \sin(\omega_s t - \frac{2\pi}{3})]^t. \quad (7)$$

then, if $\frac{2\pi}{3p}\frac{\partial x}{\partial t}$ is sufficiently smaller than ω_s, the current $\boldsymbol{I}$ is expressed as

$$\boldsymbol{I} = \frac{\partial}{\partial t}(\boldsymbol{C V_s}) = [I_m(0), I_m(\frac{2\pi}{3}), I_m(-\frac{2\pi}{3})]^t, \quad (8)$$

where

$$I_m \simeq \frac{3}{2}\omega_s V_a C_a \cos(\omega_s t + \frac{2\pi}{3p}x + \phi). \quad (9)$$

Since I_m contains information about slider position x, feeding I_m into synchronous detection gives x.

C. Control system

We fabricated levitation drive control system that can handle levitation gap and slider displacement at the same time. The system is composed of driving control part, levitation control part, and compensation part. Fig. 10(a) and (b) show the parts of driving control system and of levitation control system. Fig. 10(c) illustrates the whole block diagram of the controller. The controller is implemented on DSP system (d-space, DS1104).

In driving control part, the output of position sensor, shown in Fig. 10(a), is subtracted from the target position to give position error. The position error is fed into PD driving controller to generate thrust force command f_x^*, which is processed in three-phase signal generator to make three-phase driving signals as in (1) and (2). Finally, the three-phase driving signals are amplified by high-voltage amplifiers (Trek, Model 609C), and then applied to driving electrodes of the motor.

Basic operation of levitation control part is described as follows. The output of gap sensor is subtracted from target gap to give gap error e_d, which is fed into PD levitation

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.

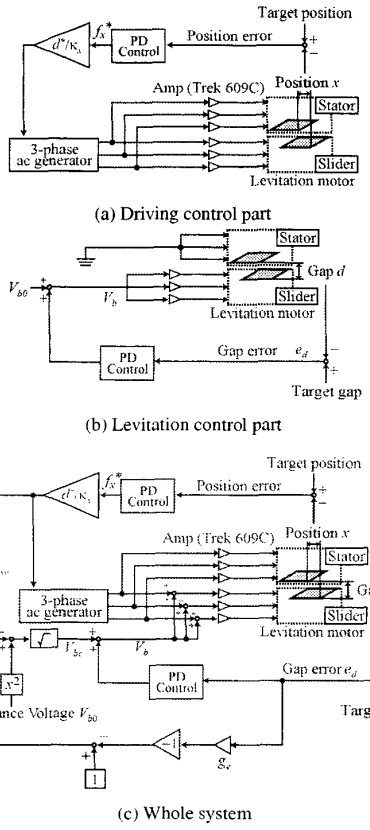


Fig. 10. Control system for electrostatic levitation motor

controller. The output of PD controller is added to balance voltage V_{b0} to make bias voltage V_b, which is applied to slider. Stator is grounded. Balance voltage V_{b0} is determined so that attractive force balances the slider mass mg at target gap d^*.

$$V_{b0} = d^* \sqrt{\frac{mg}{\kappa_b}} \tag{10}$$

Basically, the whole system is a combination of driving control part and levitation control part. However, since driving voltage influences attractive force, the whole system is not a simple merge of the two systems, but it includes a compensation part, where influence from driving voltage is feed-forwarded to levitation control part. In the whole system, bias voltage V_b is superposed on three-phase driving signal for the slider. Although in Fig. 10(b), V_b was the sum of PD controller output and balance voltage V_{b0}, it is rearranged as the sum of *compensated balance voltage* V_{bc} and PD controller output. The compensated balance voltage is calculated based on the influence from driving voltage to attractive force in the compensation part.

The attractive force f_{dl} caused by driving voltage V_0 is

$$f_{dl} = \kappa_a \frac{V_0^2}{d^2}, \tag{11}$$

and by using Taylor series, equation (11) above can be replaced as

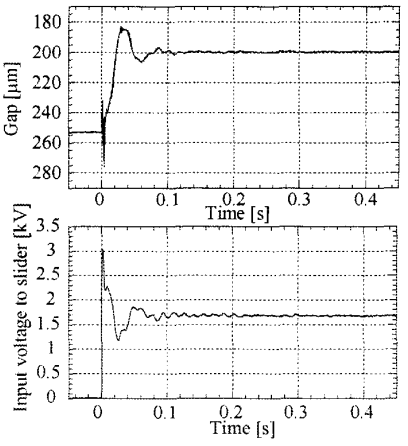


Fig. 11. Gap and voltage variations when rising a slider of electrostatic levitation motor (Driving control system was cut off); [top] Gap variation; [bottom] Voltage variation

$$f_{dl} = \kappa_a V_0^2 \left(1 - \frac{d - d^*}{2d^*}\right). \tag{12}$$

Since the compensated balance voltage V_{bc} balances slider mass at the target gap d^*, it is decided as:

$$V_{bc} = d^* \sqrt{\frac{mg - f_{dl}}{\kappa_b}} = \sqrt{V_{b0}^2 - \frac{\kappa_b}{\kappa_a}\left(1 - \frac{d - d^*}{2d^*} V_0^2\right)}. \tag{13}$$

By defining $g_{drive} = \kappa_b / \kappa_a$, $e_d = d - d^*$, and $g_e = 1/2d^*$, equation (13) is rewritten as

$$V_{bc} = \sqrt{V_{b0}^2 - g_{drive}(1 - g_e e_d) V_0^2}, \tag{14}$$

The compensation part is implemented based on (14).

VI. Experiments

We carried out levitation drive experiment in vacuum chamber. The pressure was ranging from 1×10^{-3}Pa to 5×10^{-4}Pa. For the experiment location of counter mass was adjusted so that the attractive force at balance state is 0.4N. The experiment was done with the following procedure. First, we levitated the slider using only levitation control part. At this process, driving control part and compensation part was deactivated. After levitation, we activated driving control part and compensation part to realize levitation drive.

A. Levitation process

First, we levitated the slider using levitation control part. Initially, the slider was located at a gap of 250μm supported by micrometer head. Activating levitation controller raise slider to the target gap 200μm. Experiment conditions were as follows; proportional gain: 1.8×10^7kV/m, derivative gain: 1×10^5kV·s/m, and balance voltage V_{b0}: 500V. The step response of the levitation process is shown in Fig. 11. Steady-state error was about 0.2μm.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.

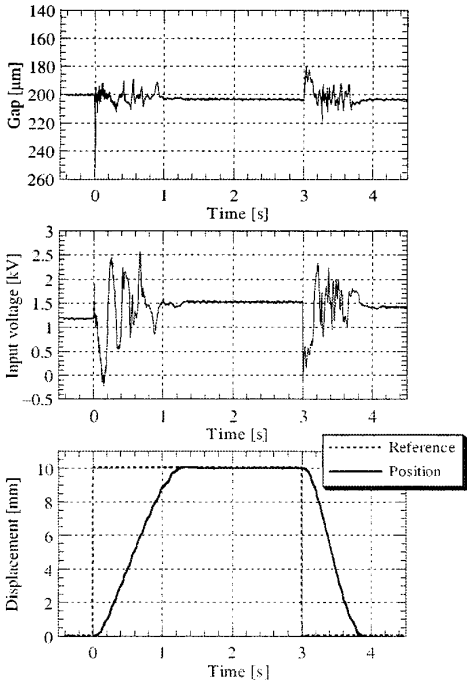


Fig. 12. 10mm step response of electrostatic levitation motor; [top] Gap variation; [middle] Voltage variation; [bottom] Position reference and slider

B. Driving performance

After levitation, we activated both the levitation system and driving system with keeping levitation controller active. The feedback gains for driving controller were as follows; proportional gain for driving: 3.2×10^2V/N, derivative gain for driving: 75V·s/N. While levitation controller was keeping the levitation gap of 200μm, we fed step input of $\pm$10mm into driving control system. The response against the step input is plotted in Fig. 12. In the displacement response, overshoot was hardly seen. Steady-state error of displacement was about 0.2mm. Levitation gap was disturbed by the driving. However, gap error due to the disturbance was kept within $\pm$20 with the help of the compensation part. Driving speed was 20 mm/s at the maximum. We think that the performance will be improved by reducing levitation gap. To reduce levitation gap, the gap variance during driving should be lowered. This will be achieved by improving compensation part of the controller.

VII. Conclusion

In this paper, the control systems and experiments of levitation motor using DEMED was described. Conclusions as below can be made.

1. By applying bias voltage to slider's electrodes, levitation and driving can be controlled using the same electrodes.
2. From the relationship between levitation gap, thrust force, and attractive force, an optimal controlling method is realized.
3. We successfully created a prototype motor that can be levitated and driven stably in vacuum condition.

In the experiments, levitation drive was successfully demonstrated. We believe this results confirm our levitation motor theory.

References

[1] H.Bleuler, "A Survey of Magnetic Levitation and Magnetic Bearing Types", JSME International Journal, Vol.5, No.3, pp.335-342 (1992)

[2] M.E.Williams, D.L.Trumper and R.Hocken, "Magnetic bearing stage for photolithography", CIRP Annals, Vol.42, No.1, pp.607-610 (1993)

[3] T.Shinshi, C.Iijima, K.Sato and A.Shimokohbe, "Precision Positioning of Magnetic Levitation System Using Hall Element", Proc. of the Fourth International Conference on Motion and Vibration Control, pp.1089-1094 (1998)

[4] S.D.Berger and J.M.Gibson, "New approach to projection-electron lithography with demonstrated 0.1μm linewidth", Applied Physics Letter, vol.57, pp.153-155 (1990)

[5] H.C.Pfeiffer and W.Stickel, "PREVAIL - An E-Beam Stepper with Variable Axis Immersion Lenses", Microelectronic Engineering, Vol.27, pp.143-146 (1995)

[6] L.Han, et al., "Performance investigation of Coulomb interaction-limited high throughput electron beam lithography based on empirical modeling", Journal of Vacuum Science and Technology B, Vol.16, No.6, pp.3215-3220 (1998)

[7] H.C.Pfeiffer, et al., "Projection reduction exposure with variable axis immersion lenses: Next generation lithography", Journal of Vacuum Science and Technology B, Vol.17, No.6, pp.2840-2846 (1999)

[8] T.Yahiro, et al., "Direct measurement of Coulomb effects in electron beam projection lithography", Journal of Vacuum Science and Technology B, Vol.19, No.6, pp.2468-2473 (2001)

[9] R.S.Dhaliwal, et al., "PREVAIL-Electron projection technology approach for next-generation lithography", IBM journal of Research and Development, Vol.45, No.5, p.615-638(2001)

[10] K.Suzuki, et al., "Nikon EB Stepper: the latest development status", Proc. of SPIE The International Society for Optical Engineering, Emerging Lithographic Technology V, Vol.4343, pp.80-87 (2001)

[11] S.Kawata, et al., "Progress and Preliminary Results on EB stepper", Journal of Applied Physics Part1, Vol.41, No.6B, pp.4136-4140 (2002)

[12] H.Fujita and A.Omodaka, "Electrostatic actuators for micro mechatronics", Proc. IEEE Micro Robots and Tele Operators Workshop, Hyannis, MA, U.S.A., Nov. 1987

[13] S.F.Bart, et al., "Design considerations for micromachined electric actuators", Sensors and Actuators, Vol.14, pp.269-292 (1988)

[14] L.S.Fan, Y.C.Tai and R.S.Muller, "IC-processed electrostatic micromotors", Sensor and Actuators, Vol.20, pp.41-48 (1989)

[15] W.Trimmer and R.Jebens, "An Operational Harmonic Electrostatic Motor", Proc. of IEEE Micro Electro Mechanical Systems Workshop, pp.13-16 (1989)

[16] S.C.Jacobsen, et al., "The Wobble Motor: An Electrostatic, Planetary-Armature, Micfro-actuator", Proc. IEEE Micro Electro Mechanical Systems Workshop, pp.17-24 (1989)

[17] H.Fujita, et al., "Electrostatic Micro Actuators", Tech Digest of 7th Sensor Symposium, Tokyo, pp.169-172 (1988)

[18] T.Niino, T.Higuchi, and S.Egawa, "Dual Excitation Multiphase Electrostatic Drive", Conf. Record of 1995 IEEE IAS annual meeting, pp. 1318-1325 (1995)

[19] T.Niino, S.Egawa, and T.Higuchi, "High-power and High efficiency Electrostatic Actuator", Proc.1993 IEEE Workshop on Micro Electro Mechanical System, pp.236-241 (1993)

[20] H.W.knoebel, "The Electric Vacuum Gyro", Control Engineering, Vol. 11, p70, (1964)

[21] S.kumar, D.cho, and W.Carr, "A proposal for electrically levitating micromotors", Sensors and Actuators A, Vol.24, pp.141-149 (1990)

[22] A.Yamamoto, T.Niino, and T.Higuchi, "High Power Electrostatic Actuator with Novel Electrode Design", Proc.IEEE MEMS'98, pp.408-413 (1998)

[23] A.Yamamoto, T.Niino, and T.Higuchi, "A High-Power Electrostatic Linear Servo Motor", Proc. LDIA'98, pp. 212-215 (1998)

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:07:21 UTC from IEEE Xplore. Restrictions apply.